

CPV301 - Lab1

Color Models and Color Spaces (OpenCV & Google Colab) vubc

General Information (for All Groups)

Platform: Google Colab (Python + OpenCV)

Group work: Each group (Group 1–7) receives a slightly different version of the lab.

Objectives

By the end of this lab, students will be able to:

- Understand and distinguish **color models** vs **color spaces**.
- Describe and use common color spaces: RGB/BGR, HSV/HSL, Lab, YCrCb, CIE XYZ/Lab/Luv, CMYK, YUV/YCbCr, opponent, LMS.
- Use OpenCV in Google Colab to convert images between different color spaces and observe the differences.
- Choose appropriate color spaces for different computer vision tasks (segmentation, color difference, video/compression, etc.).

Instructions

- Work in **Google Colab** only.
- Each group creates one notebook and names it:

`CV_Lab_ColorSpace_GroupX.ipynb`

where X is the group number (1–7).

- The notebook must include:
 - A short introduction and objectives (in English).
 - Clear section headings for each task.
 - All code cells and their outputs (plots, images, tables).
 - Written answers and comments in Markdown cells.
- At the end, run all cells and export:
 - The notebook file: `.ipynb`.
 - A PDF version of the notebook (via browser “Print to PDF”).
- Submission:
 - Submit both `CV_Lab_ColorSpace_GroupX.ipynb` and `CV_Lab_ColorSpace_GroupX.pdf` to edunext platform.
- Use at least 1–2 color images with diverse colors (outdoor scenes, people, objects, etc.).

Report Structure (Recommended for All Groups)

Each group should prepare a short report (3–6 pages) inside the Colab notebook, following this structure:

1. Introduction (objectives, short theory summary).
2. Experimental setup (environment, important code, images used).
3. Results (figures, plots, tables, comments).
4. Answers to the theory questions for your group.
5. Conclusion (what you learned about choosing color spaces).

Group 1 – Lab Specification

Part A – Theory Questions

Answer the following questions in Markdown cells in your Colab notebook:

1. Define **color model** and **color space**. Give at least two examples for each.
2. Explain the difference between “RGB” as a color model and “sRGB” as a specific color space.
3. Describe why HSV/HSL can be more convenient than RGB for color thresholding tasks.

Part B – Practical Tasks (OpenCV in Colab)

1. Read a color image using OpenCV (BGR).
2. Convert the image to the following color spaces:
 - RGB
 - HSV
 - HLS
 - Lab
 - YCrCb
3. Display the six images (original BGR, RGB, HSV, HLS, Lab, YCrCb) in a 2×3 grid using `matplotlib`.
4. Plot the histogram of the **H channel** in HSV and the **L channel** in Lab. Compare them and comment:
 - Which channel is more suitable for analyzing brightness / lightness? Why?
5. Select two pixels with clearly different colors (you can hard-code coordinates).
 - Compute the Euclidean distance between their colors in RGB.
 - Convert the image to Lab and compute the Euclidean distance between their colors in Lab.
 - Comment: which distance seems more consistent with human perception?

Group 2 – Lab Specification

Part A – Theory Questions

1. Explain which type of color model RGB and CMYK represent (additive vs. subtractive).
2. Describe the meaning of the three channels in YCrCb.
3. Give one application where using Lab is more advantageous than RGB.

Part B – Practical Tasks

1. Read a color image (BGR).
2. Convert the image to:
 - RGB
 - HSV
 - Lab
 - YCrCb
 - XYZ

3. Display each channel of YCrCb (Y, Cr, Cb) as separate grayscale images.
 - Comment: which channel is most similar to a grayscale intensity image?
 4. Create grayscale images in two ways:
 - (a) Use `cv2.cvtColor(img_bgr, cv2.COLOR_BGR2GRAY)`.
 - (b) Take the Y channel from YCrCb and normalize it if needed.
 - Compare the results and explain any visible differences.
 5. Choose two small patches (e.g., 10×10 pixels each) from the image:
 - Compute the average color of each patch in RGB and in Lab.
 - Compute the Euclidean distance between the two patches in RGB and in Lab.
 - Comment on which distance seems to better reflect the perceived color difference.
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Group 3 – Lab Specification

Part A – Theory Questions

1. Briefly define CIE XYZ and its role in defining other color spaces.
2. Compare CIE Lab and CIE Luv in terms of:
 - Similarities,
 - Differences,
 - Typical application contexts.
3. Explain the basic idea of the ΔE (CIE76) color difference measure in Lab (only the concept, not the detailed formula).

Part B – Practical Tasks

1. Read a color image (BGR).
 2. Convert the image to:
 - XYZ
 - Lab
 - Luv
 3. Select five pairs of sample pixels (by coordinates) with varying degrees of similarity/difference in color.
 4. For each pair, compute:
 - Euclidean distance in RGB.
 - Euclidean distance in Lab (treat this as a simple ΔE CIE76).
 5. Create a table in your notebook that includes: pair index, RGB distance, Lab distance, and subjective perception (your group's opinion).
 - Comment: in which cases does Lab distance better align with human perception than RGB distance?
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Group 4 – Lab Specification

Part A – Theory Questions

1. Explain the concept of the **gamut** of a color space.
2. Compare (at a conceptual level) sRGB and Adobe RGB in terms of gamut and typical applications.
3. Give one reason why understanding gamut is important in imaging and display.

Part B – Practical Tasks

1. Choose one image with a rich variety of colors (including highly saturated regions).
 2. Read the image with OpenCV (assume it is in sRGB).
 3. Convert it to HSV and:
 - Randomly sample a subset of pixels (e.g., 5,000–10,000 pixels).
 - Plot a scatter plot of their (H, S) values to visualize the color distribution.
 4. Generate a simple Hue–Saturation grid:
 - Fix V (e.g., V is constant) and vary H and S over a grid.
 - Convert this synthetic HSV image to RGB and display it.
 - This should give intuition about how the HSV space “fills” the gamut.
 5. In the report, include at least one diagram (copied and redrawn or referenced) that illustrates the difference between sRGB and Adobe RGB gamuts, and relate it to your HSV scatter plot observations.
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Group 5 – Lab Specification

Part A – Theory Questions

1. Describe the main idea of an **opponent color space** (e.g., red–green, yellow–blue, light–dark).
2. Explain the relationship between opponent color spaces and the human visual system.
3. Give an example of a computer vision task that can benefit from using an opponent color space.

Part B – Practical Tasks

1. Read a color image (BGR) and convert it to RGB.
2. Implement a simple opponent color transform, for example:

$$O_1 = \frac{R - G}{\sqrt{2}}, \quad O_2 = \frac{R + G - 2B}{\sqrt{6}}, \quad O_3 = \frac{R + G + B}{\sqrt{3}}.$$

3. Normalize each channel O_1, O_2, O_3 to $[0, 255]$ and display them as grayscale images.
 4. Comment:
 - Which channel best separates red vs. green regions?
 - Which channel best captures overall intensity?
 5. Convert the same image to HSV and display the H and S channels.
 - Compare how well the opponent channels vs. HSV channels separate different colored regions.
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Group 6 – Lab Specification

Part A – Theory Questions

1. Explain the role of LMS in modeling the human visual system (three cone types L, M, S).
2. Describe the concept of **color constancy** and why it is important in computer vision.
3. Briefly outline one method to achieve color constancy (e.g., Gray World, White Patch, or Retinex).

Part B – Practical Tasks

1. Use 2–3 images of the same scene under different lighting conditions (you can capture them or use a dataset).
 2. Read each image with OpenCV and convert it to a suitable color space (e.g., RGB and Lab).
 3. Implement a simple Gray World white balancing algorithm:
 - Compute the mean of each RGB channel.
 - Compute scaling factors to map these means to a neutral gray level.
 - Apply the scaling to each channel and clip values to $[0, 255]$.
 4. Apply the algorithm to all images and compare:
 - Before vs. after white balancing.
 - Across different lighting conditions.
 5. Optionally, experiment with working in Lab (e.g., adjusting only the L channel) and comment on any differences in the results.
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Group 7 – Lab Specification

Part A – Theory Questions

1. Distinguish between YUV and YCbCr (historical origin and modern usage).
2. Explain why, in image/video compression (JPEG, MPEG, etc.), chroma channels (Cb, Cr or U, V) are often subsampled.
3. Give one example of a computer vision task where separating luma/chroma makes processing easier.

Part B – Practical Tasks

1. Take one frame from a video or use a single image as a representative frame.
2. Convert the frame to:
 - YUV
 - YCrCb
3. Display the following channels as grayscale images:
 - Y (from YUV or YCrCb),
 - U and V (or Cr and Cb).

Comment on how these channels relate visually to the original image.
4. Perform a simple segmentation experiment:
 - Threshold on the Y channel to separate bright vs. dark regions.
 - Threshold on chroma channels (e.g., Cr/Cb) to roughly highlight a region with a specific color (e.g., skin-like or a colored background).
5. Compare segmentation results when thresholding:
 - Directly on RGB channels.
 - On Y and chroma channels.

Comment on which approach is more robust or intuitive.